

Observation Task 2: Object-Oriented Programming in Java

Aim / Objective

To understand and implement the core Object-Oriented Programming concepts in Java by developing a simple Library Management System.

Description

The Library Management System is a simple Java program that demonstrates classes and objects, constructors, encapsulation, inheritance, method overloading, method overriding, polymorphism, abstraction, and interfaces using library-related operations.

System Requirements

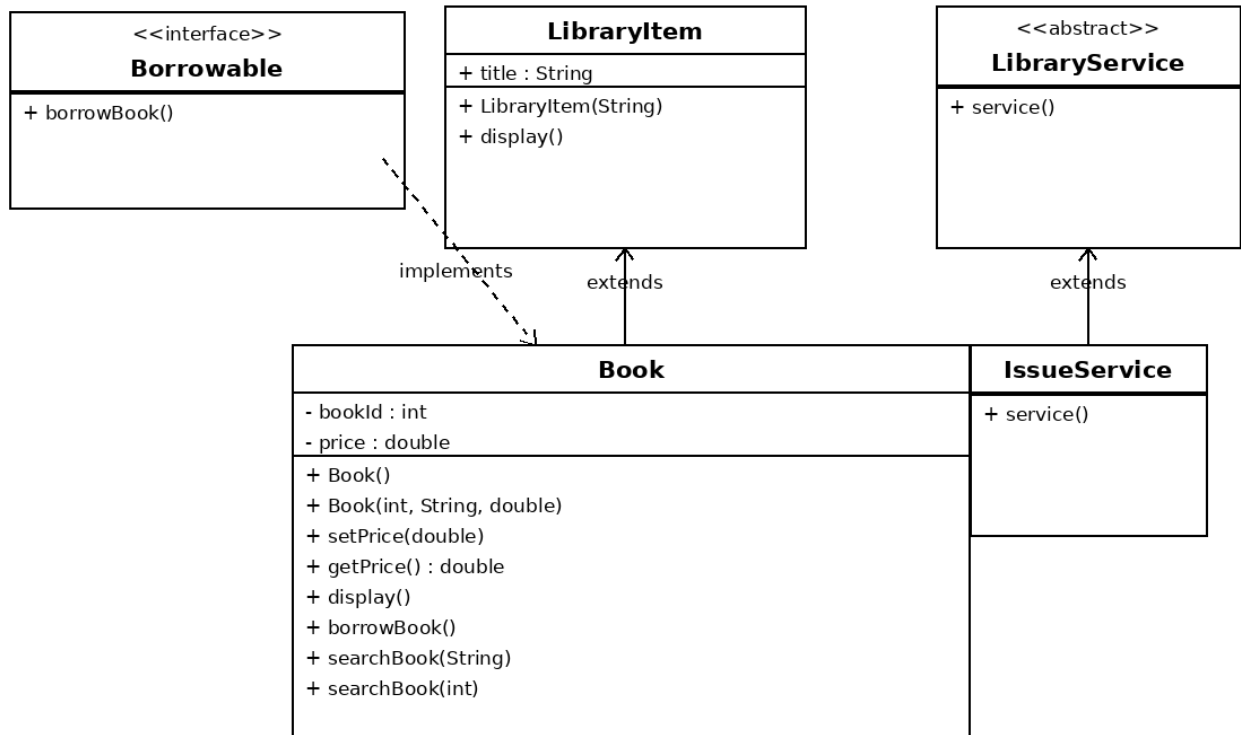
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm

1. Start the program.
2. Create the Borrowable interface with borrowBook().
3. Create LibraryItem with a title and display() method.
4. Create Book by extending LibraryItem and implementing Borrowable.
5. Use default and parameterized constructors to initialize Book objects.
6. Keep book price private and access it through setPrice() and getPrice().
7. Override display() and overload searchBook() with String and int parameters.
8. Create abstract LibraryService and implement it using IssueService.
9. Create a Book object and call the required methods from main().
10. Display the results and stop the program.

UML Diagram



Source Code

```

// Interface
interface Borrowable {
    void borrowBook();
}

// Base class
class LibraryItem {
    String title;

    LibraryItem(String title) {
        this.title = title;
    }

    void display() {
        System.out.println("Library Item: " + title);
    }
}

// Inheritance + Method Overriding
class Book extends LibraryItem implements Borrowable {
    private int bookId;
    private double price;

    // Default constructor
    Book() {
        super("Unknown");
        bookId = 0;
        price = 0;
    }

    // Parameterized constructor
    Book(int bookId, String title, double price) {
        super(title);
    }
}

```

```

        this.bookId = bookId;
        this.price = price;
    }

    // Encapsulation
    void setPrice(double price) {
        this.price = price;
    }

    double getPrice() {
        return price;
    }

    @Override
    void display() {
        System.out.println("Book ID : " + bookId);
        System.out.println("Title   : " + title);
        System.out.println("Price   : " + price);
    }

    public void borrowBook() {
        System.out.println("Book borrowed successfully.");
    }

    // Method Overloading
    void searchBook(String title) {
        System.out.println("Searching book: " + title);
    }

    void searchBook(int bookId) {
        System.out.println("Searching book ID: " + bookId);
    }
}

// Abstraction
abstract class LibraryService {
    abstract void service();
}

class IssueService extends LibraryService {
    void service() {
        System.out.println("Book issue service");
    }
}

public class LibraryManagementSystem {
    public static void main(String[] args) {

        // Class and Object
        Book book = new Book(101, "Java Programming", 450);

        book.display();

        // Encapsulation
        book.setPrice(500);
        System.out.println("Updated Price: " + book.getPrice());

        // Method Overloading
        book.searchBook("Java Programming");
        book.searchBook(101);

        // Interface
        book.borrowBook();

        // Abstraction
        LibraryService service = new IssueService();
        service.service();
    }
}

```

```
}  
}
```

Compilation and Execution Commands

Save as: LibraryManagementSystem.java
Compile: javac LibraryManagementSystem.java
Execute: java LibraryManagementSystem

Output Screens

Possible Output – Library Book and OOP Operations

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
  
PS C:\Users\student\Desktop\Java-Projects> javac LibraryManagementSystem.java  
PS C:\Users\student\Desktop\Java-Projects> java LibraryManagementSystem  
  
Book ID : 101  
Title   : Java Programming  
Price   : 450.0  
Updated Price: 500.0  
Searching book: Java Programming  
Searching book ID: 101  
Book borrowed successfully.  
Book issue service
```

Possible Output – Library Operations

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
  
PS C:\Users\student\Desktop\Java-Projects> java LibraryManagementSystem  
  
Book ID : 101  
Title   : Java Programming  
Price   : 450.0  
Updated Price: 500.0  
Searching book: Java Programming  
Searching book ID: 101  
Book borrowed successfully.  
Book issue service
```

Possible Output – Complete Execution

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS  
  
PS C:\Users\student\Desktop\Java-Projects> java LibraryManagementSystem  
  
Book ID : 101  
Title   : Java Programming  
Price   : 450.0  
Updated Price: 500.0  
Searching book: Java Programming  
Searching book ID: 101  
Book borrowed successfully.  
Book issue service
```

Result

Thus, the Library Management System was implemented to demonstrate the basic Object-Oriented Programming concepts of Java using classes, objects, constructors, encapsulation, inheritance, polymorphism, abstraction, and interfaces.